Filter Wizard

AAF_AFW_19k_100k

Created on 06/12/2026

ANALOG DEVICES

Filter Wizard Design Report

Filter Requirements for Low-Pass, 3rd order Butterworth

Specifications: Optimize: Noise; +Vs: 5; -Vs: 0

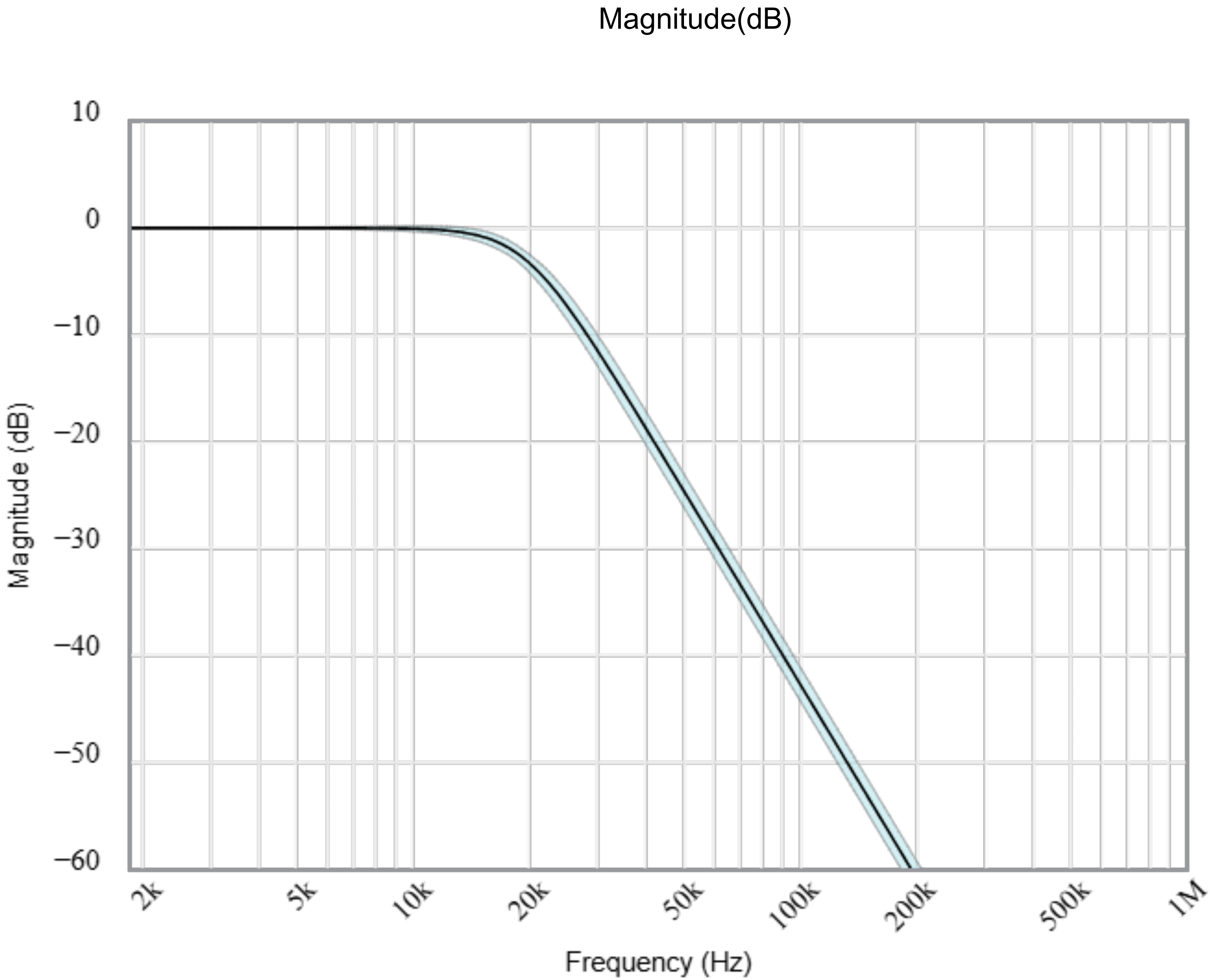
Gain: 0 dB

Passband: -3dB at 19kHz

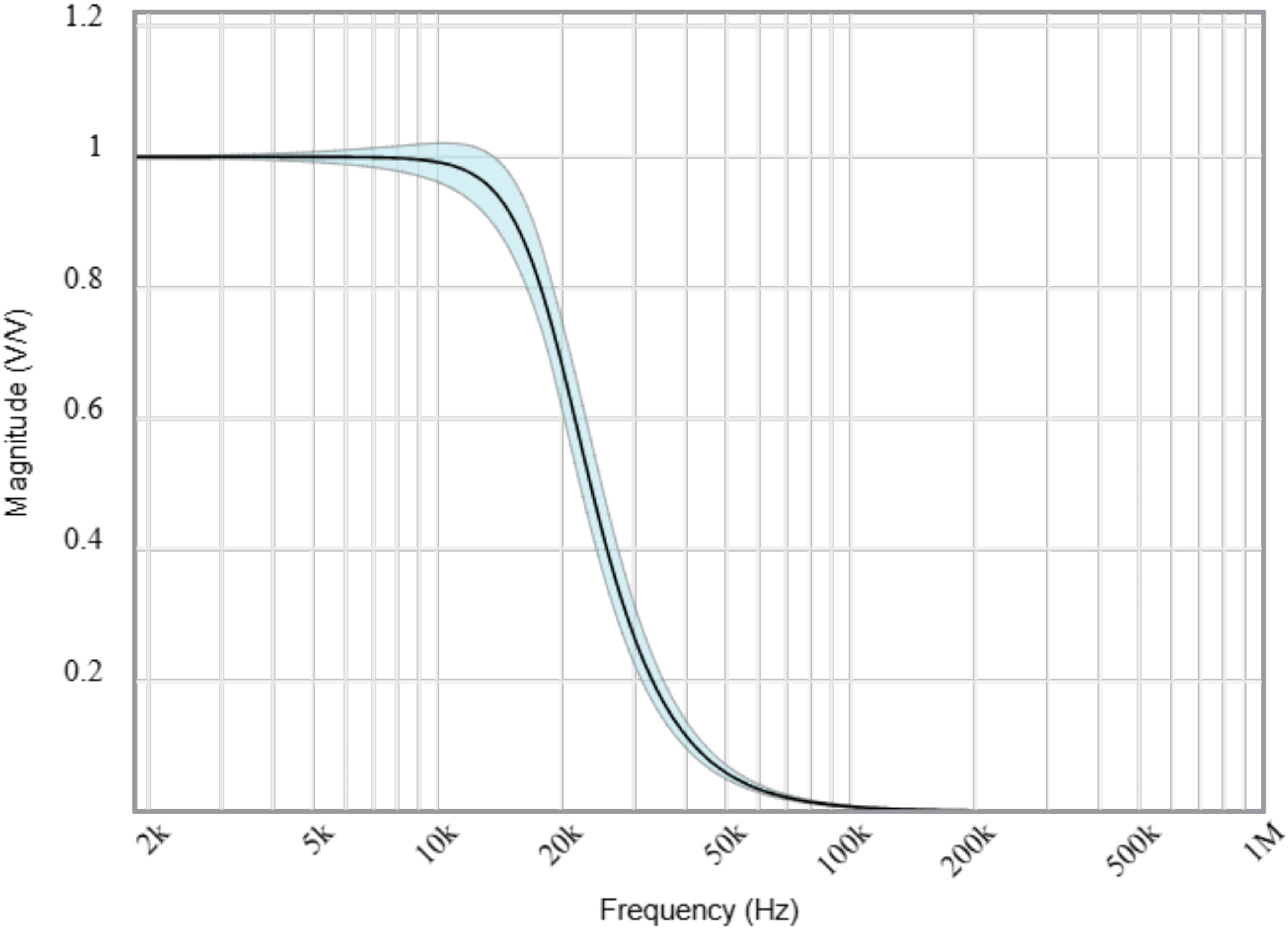
Stopband: -40dB at 100kHz

Component Tolerances: Capacitor = 5%; Resistor = 1%; Inductor = 5%; Op Amp GBW = 20%

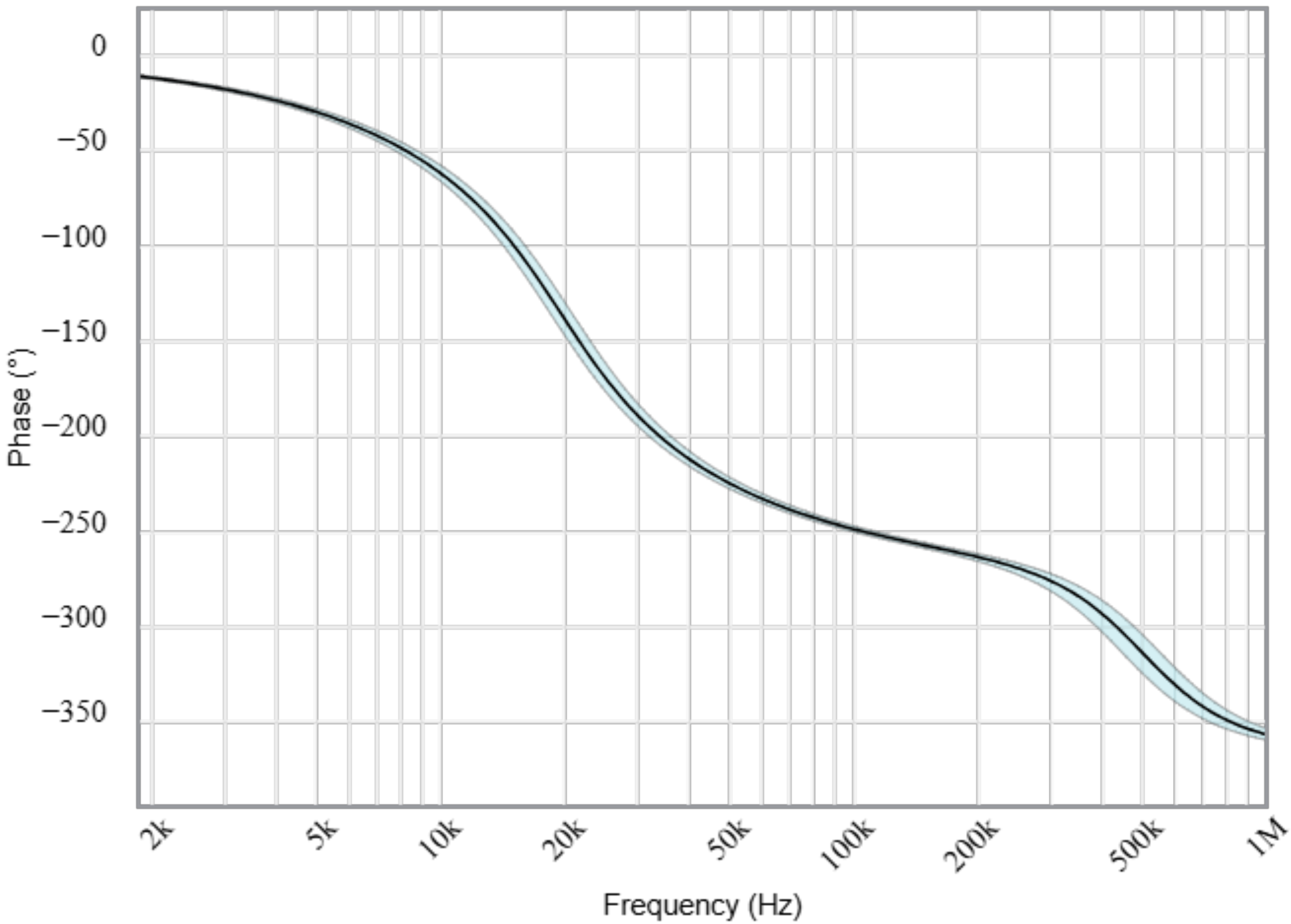
BOM: refer to BOM.csv file


Magnitude(dB)
Magnitude (dB)
10
0
−10
−20
−30
−40
−50
−60
2k
5k
10k
20k
50k
100k
200k
500k
1M
Frequency (Hz)

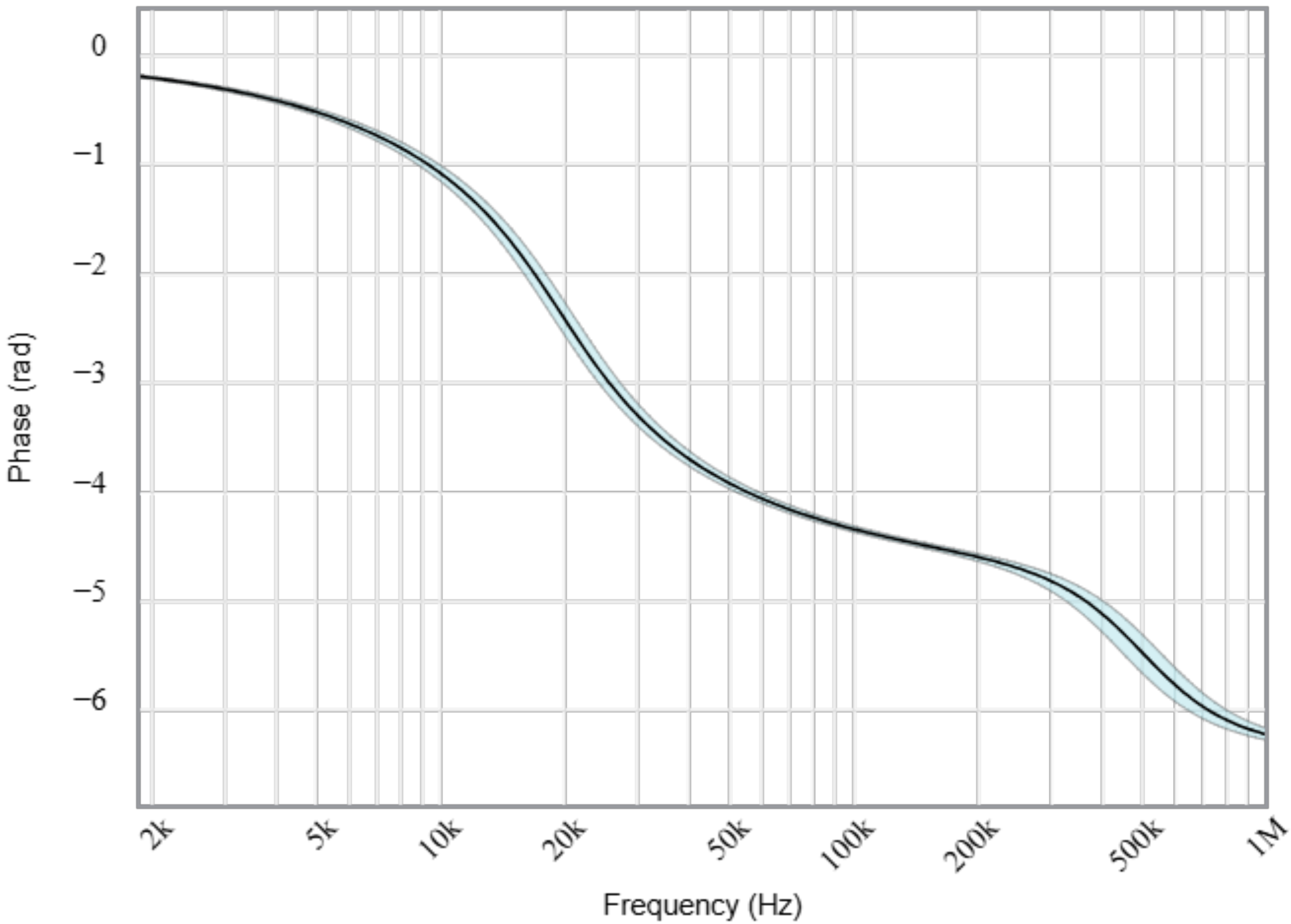
Magnitude(Volts per Volt)

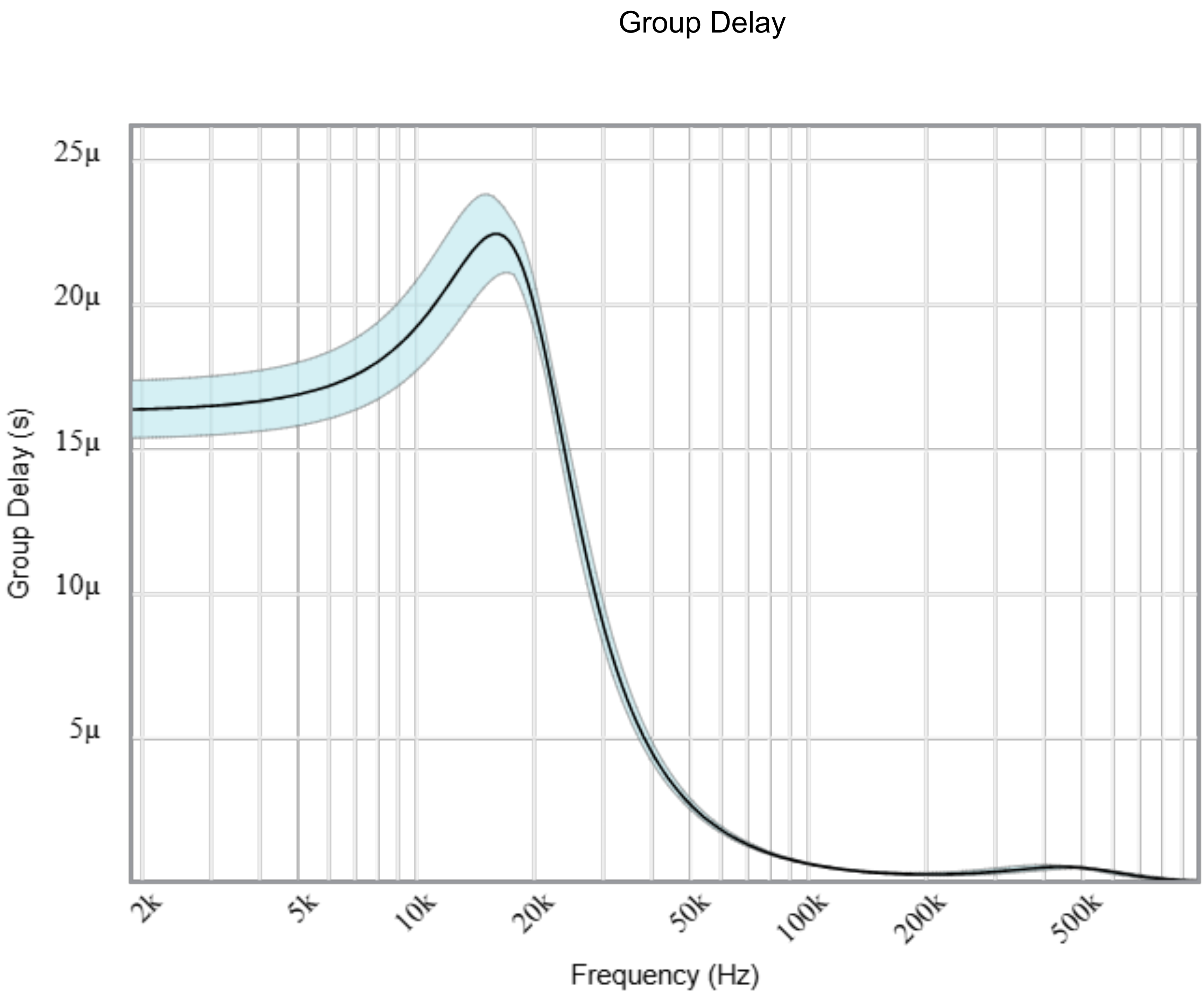


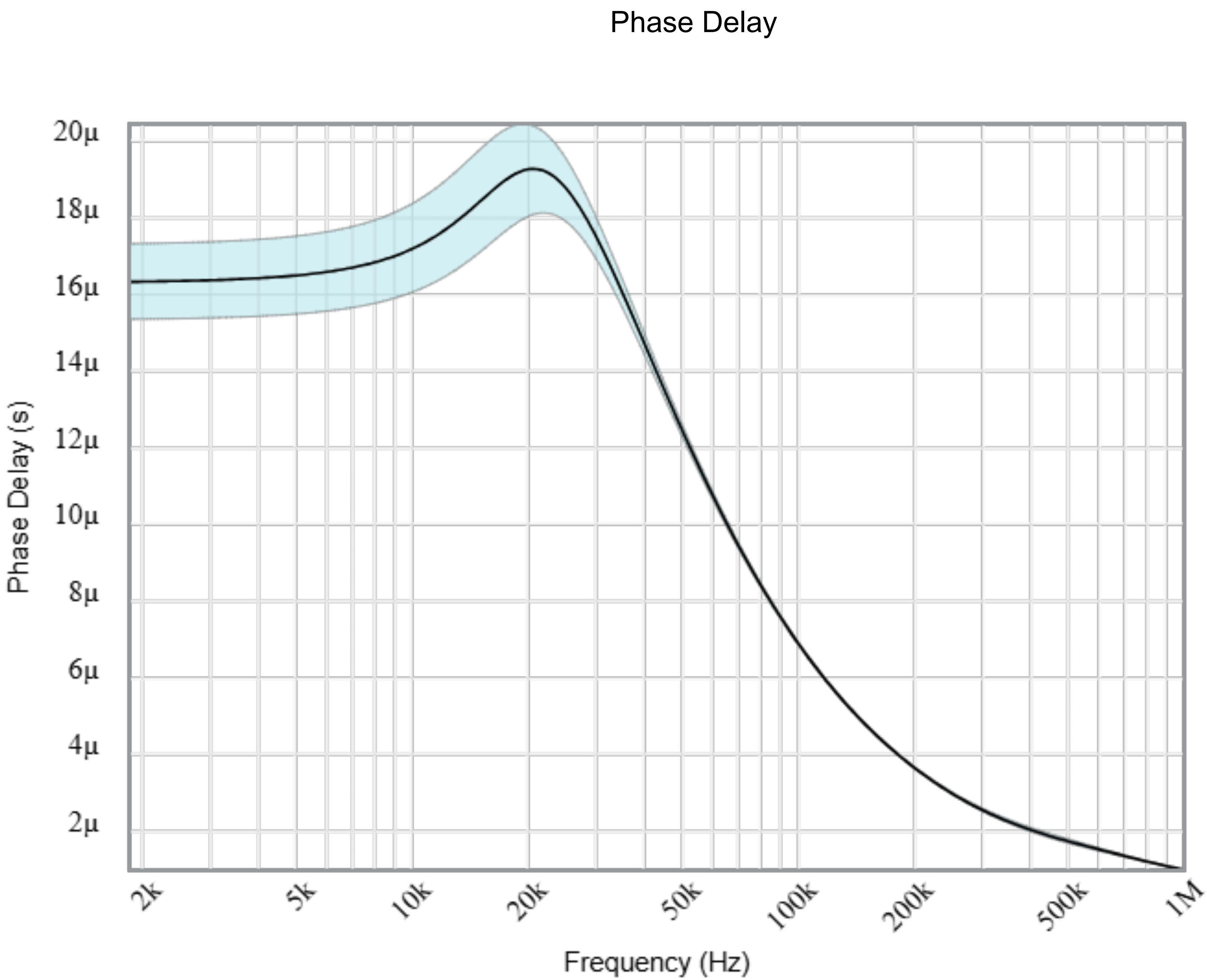
Phase(degrees)



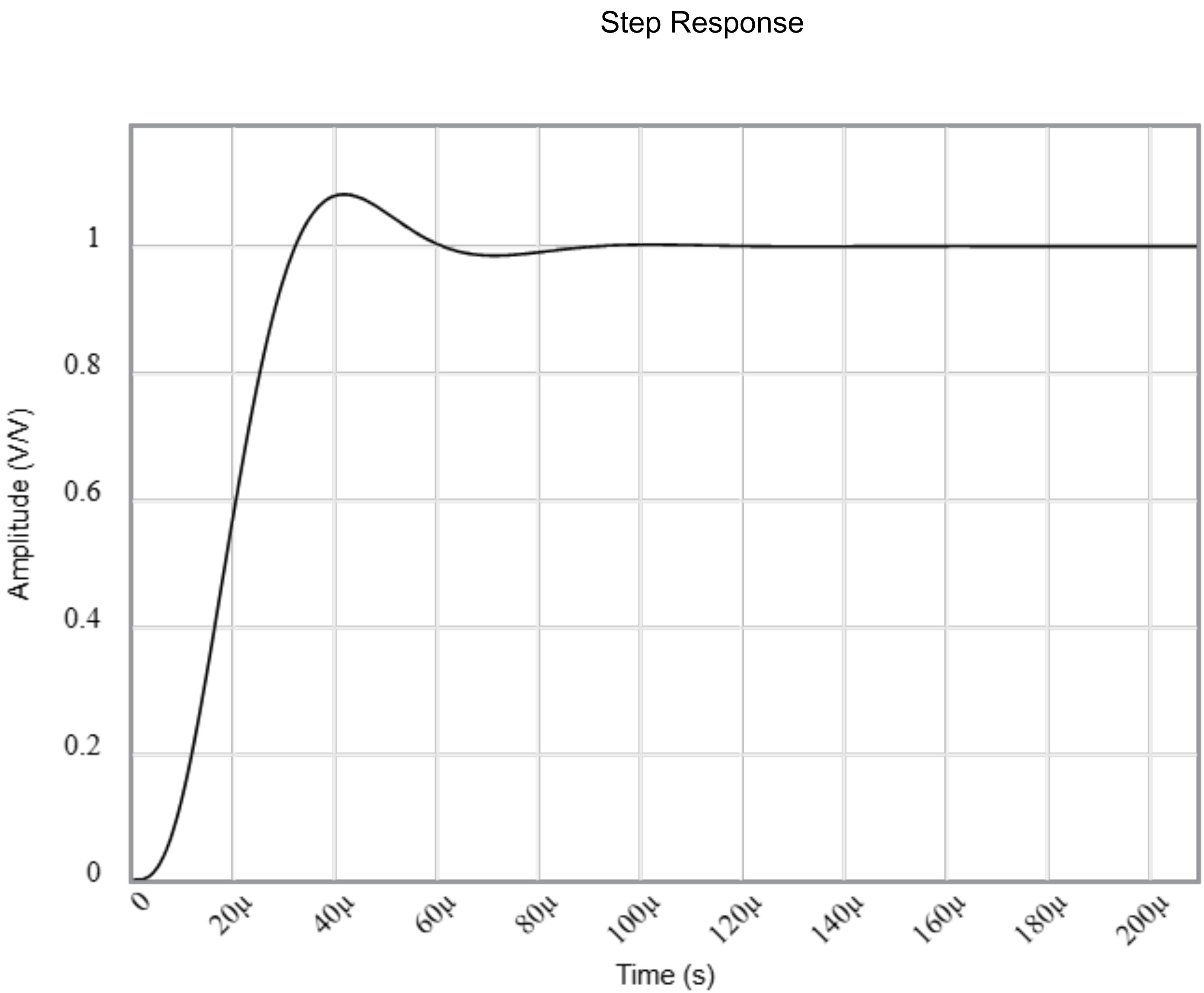
Phase(radians)



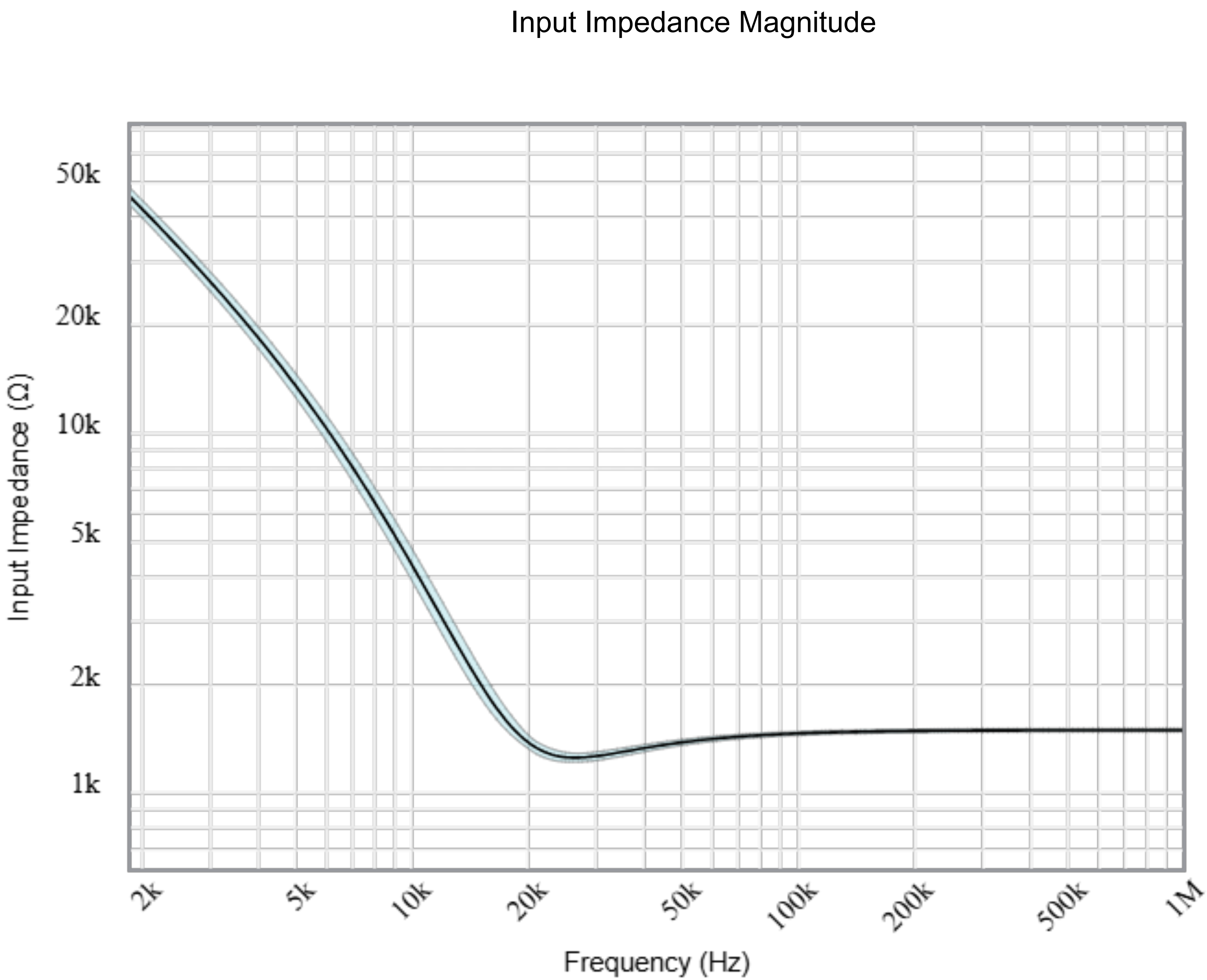

Group Delay
Group Delay (s)
25µ
20µ
15µ
10µ
5µ
2k
5k
10k
20k
50k
100k
200k
500k
Frequency (Hz)



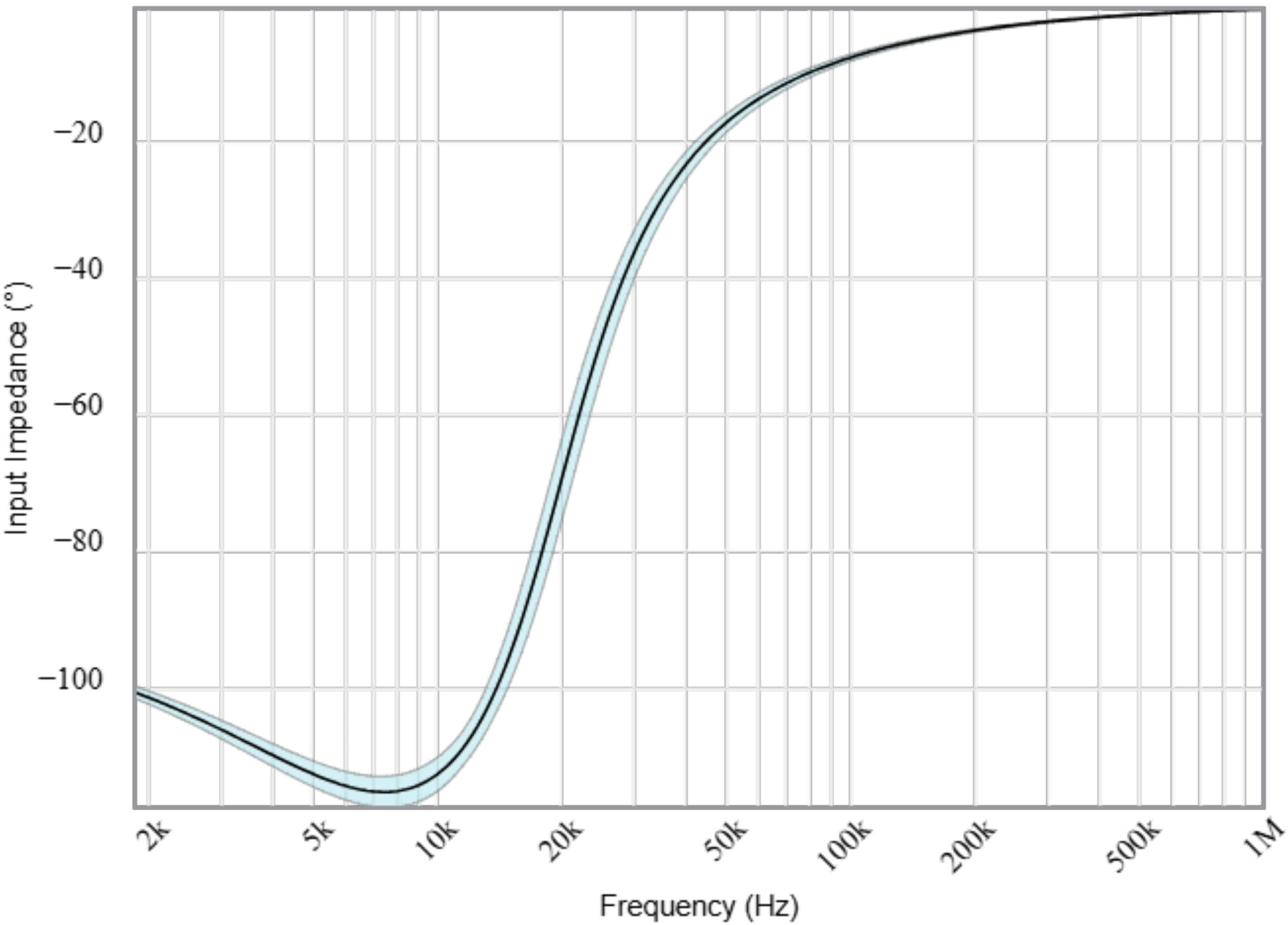
Phase Delay
Phase Delay (s)
20µ
18µ
16µ
14µ
12µ
10µ
8µ
6µ
4µ
2µ
2k
5k
10k
20k
50k
100k
200k
500k
1M
Frequency (Hz)



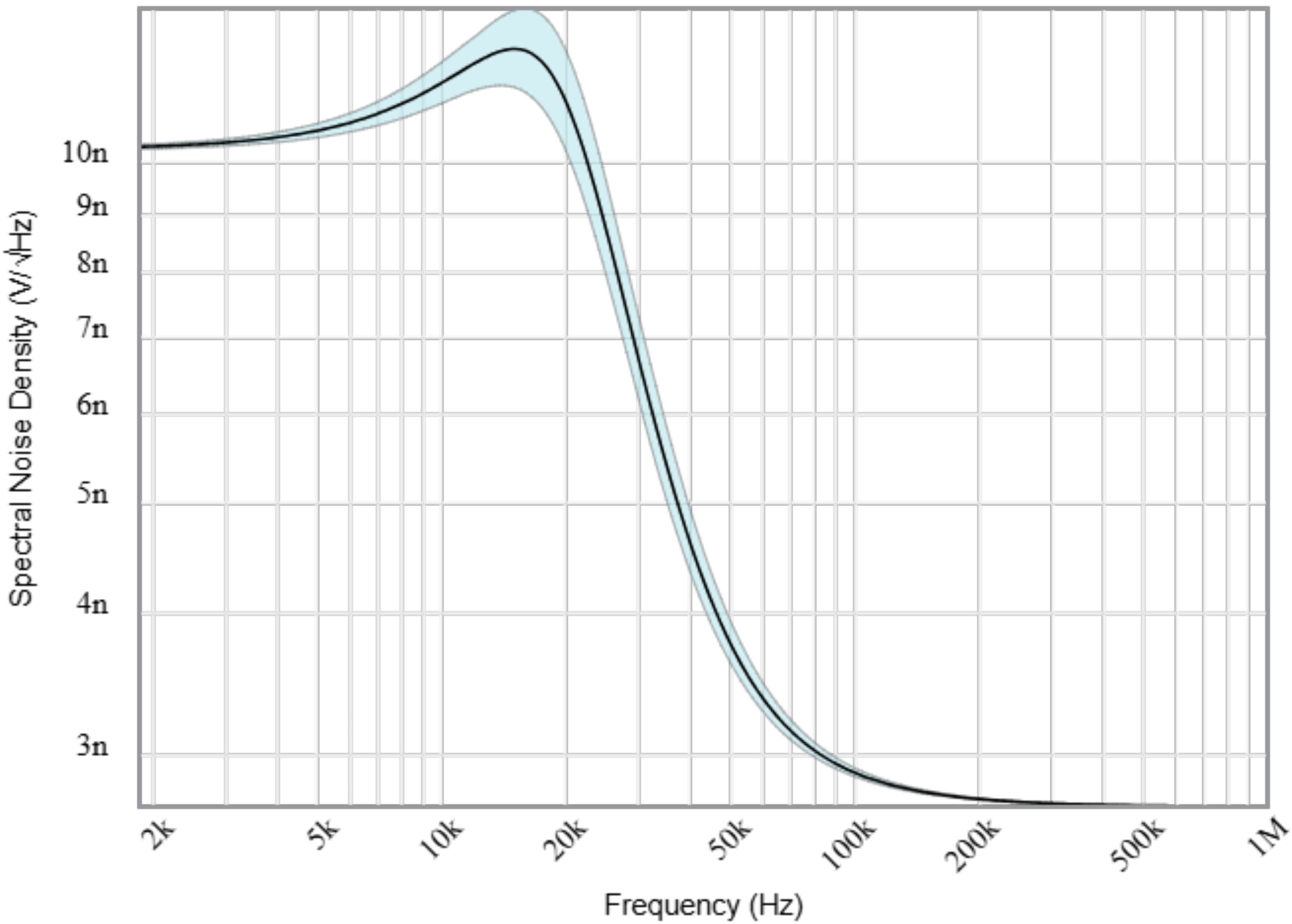
Step Response
Amplitude (V/V)
0
0.2
0.4
0.6
0.8
1
0
20μ
40μ
60μ
80μ
100μ
120μ
140μ
160μ
180μ
200μ
Time (s)


Input Impedance Magnitude
50k
20k
10k
5k
2k
1k
Input Impedance (Ω)
2k
5k
10k
20k
50k
100k
200k
500k
1M
Frequency (Hz)

Input Impedance Phase



Noise

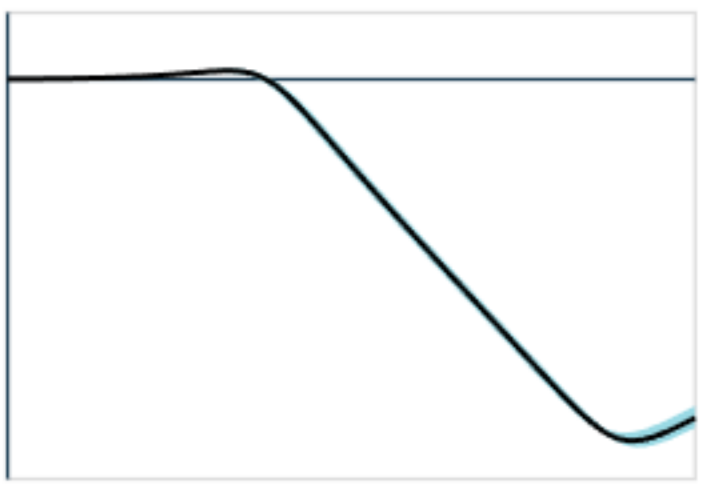


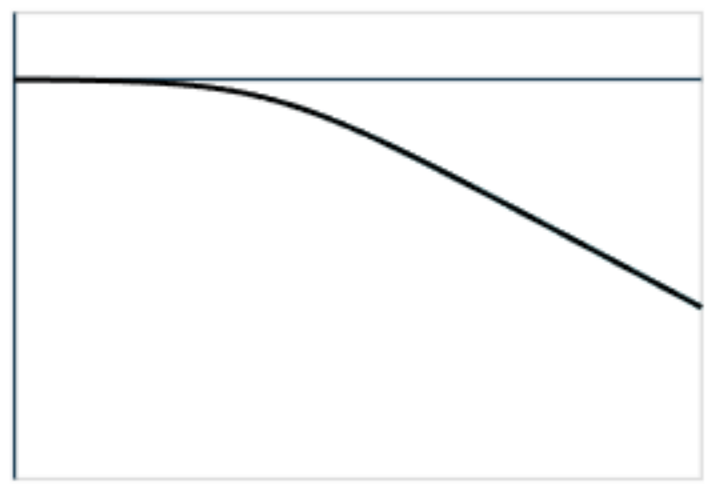
Spectral Noise Density (V/√Hz)
10n
9n
8n
7n
6n
5n
4n
3n
2k
5k
10k
20k
50k
100k
200k
500k
1M
Frequency (Hz)

Stages

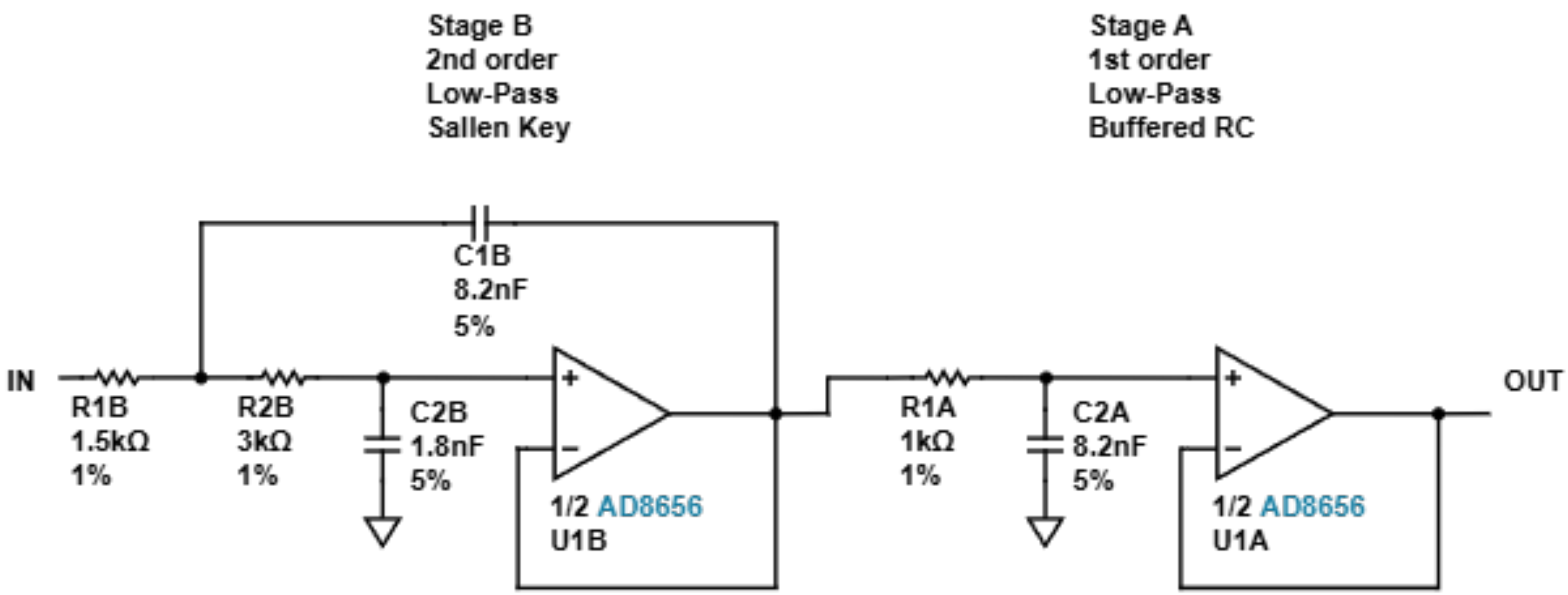
Your filter requires 2 op amp stage(s) with the following characteristics

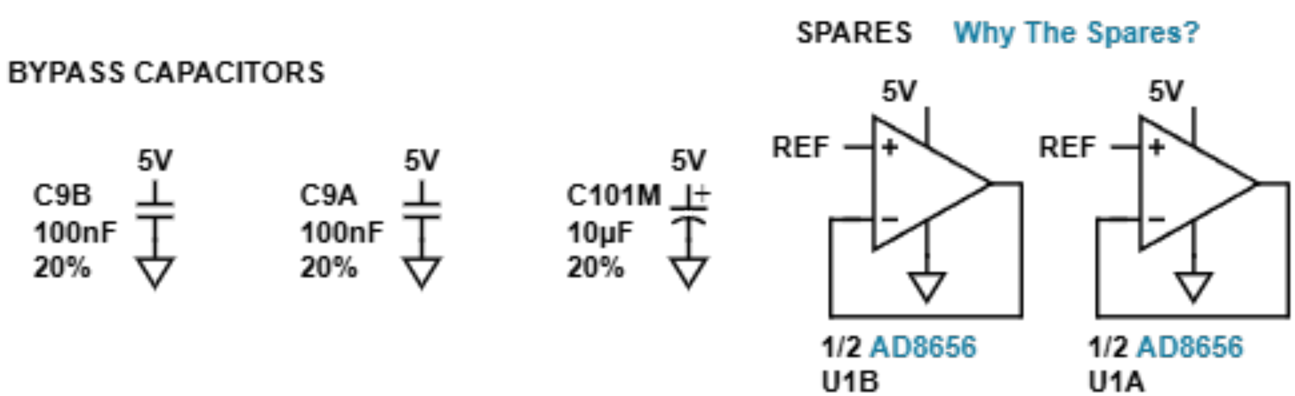
	B — 2nd order Low-Pass Sallen Key		A — 1st order Low-Pass Buffered RC	
	Target	**Simulated**	**Target**	**Simulated**
Gain (V/V):	1	1 to 1	1	1 to 1
f_p (Hz):	19k	18.4k to 20.8k	19k	18.3k to 20.6k
Q:	1	954m to 1.06	N/A	N/A to N/A





Circuit


Stage B
2nd order
Low-Pass
Sallen Key
Stage A
1st order
Low-Pass
Buffered RC
C1B
8.2nF
5%
IN
R1B
1.5kΩ
1%
R2B
3kΩ
1%
C2B
1.8nF
5%
1/2 AD8656
U1B
R1A
1kΩ
1%
C2A
8.2nF
5%
1/2 AD8656
U1A
OUT


BYPASS CAPACITORS
C9B
100nF
20%
5V
C9A
100nF
20%
5V
C101M
10µF
20%
5V
SPARES
Why The Spares?
5V
REF
1/2 AD8656
U1B
5V
REF
1/2 AD8656
U1A